

7. Exponential Distribution — Solved Problems in Data Science

1. What is Exponential Distribution?

The **Exponential Distribution** is a continuous probability distribution commonly used to model the **waiting time until the next event occurs**.

If the event rate is λ , we write:

$$X \sim \text{Exponential}(\lambda)$$

where:

- X = waiting time
- λ = average event rate
- $e \approx 2.71828$

The **Probability Density Function (PDF)** is:

$$f(x) = \lambda e^{-\lambda x}, x \geq 0$$

and:

$$f(x) = 0, x < 0$$

2. Important Formulas

Probability that waiting time is less than x

The Cumulative Distribution Function (CDF) is:

$$P(X \leq x) = 1 - e^{-\lambda x}$$

Probability that waiting time is greater than x

$$P(X > x) = e^{-\lambda x}$$

Probability between two values

$$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$$

Mean

$$E(X) = \frac{1}{\lambda}$$

Variance

$$\text{Var}(X) = \frac{1}{\lambda^2} \quad \boxed{\text{Var}(X) = \frac{1}{\lambda^2}}$$

Standard deviation

$$\sigma = \frac{1}{\lambda} \quad \boxed{\sigma = \frac{1}{\lambda}}$$

So, for an Exponential distribution:

$$\text{Mean} = \text{Standard Deviation} = \frac{1}{\lambda} \quad \boxed{\text{Mean} = \text{Standard Deviation} = \frac{1}{\lambda}}$$

Problem 1: Website Visitor Arrival

A website receives an average of **6 visitors per minute**.

Assume the time between consecutive visitors follows an Exponential distribution.

Find the probability that the next visitor arrives within **10 seconds**.

Step 1: Convert the time

Since the rate is given per minute:

$$10 \text{ seconds} = \frac{10}{60} = \frac{1}{6} \text{ minute} \quad 10 \text{ seconds} = \frac{10}{60} = \frac{1}{6} \text{ minute}$$

Therefore:

$$\lambda = 6 \quad \lambda = \frac{1}{\frac{1}{6}}$$

and:

$$\lambda = 6 \quad \lambda = 6$$

Step 2: Use the CDF

$$P(X \leq x) = 1 - e^{-\lambda x} \quad P(X \leq x) = 1 - e^{-\lambda x}$$

Therefore:

$$P(X \leq \frac{1}{6}) = 1 - e^{-6(\frac{1}{6})} = 1 - e^{-1} = 1 - e^{-1}$$

Since:

$$e^{-1} \approx 0.3679 \quad e^{-1} \approx 0.3679$$

we get:

$$P(X \leq 16) = 1 - 0.3679 P(X \leq \frac{16}{10}) = 1 - 0.3679 P(X \leq 1.6 \text{ sec}) \approx 0.6321 \boxed{P(X \leq 1.6 \text{ sec}) \approx 0.6321}$$

Answer

$$63.21\% \boxed{63.21\%}$$

Data Science Interpretation

There is approximately a **63.21% probability that the next visitor arrives within 10 seconds.**

Problem 2: API Request Waiting Time

An API receives an average of **12 requests per second.**

What is the probability that the next request arrives within **0.1 seconds?**

Given

$$\lambda = 12 \text{ } \lambda x = 12 \times 0.1 = 1.2$$

Using:

$$P(X \leq x) = 1 - e^{-\lambda x} P(X \leq 0.1) = 1 - e^{-1.2}$$

we get:

$$P(X \leq 0.1) = 1 - e^{-1.2} P(X \leq 0.1) = 1 - e^{-1.2} = 1 - 0.3010 = 0.6990$$

Since:

$$e^{-1.2} \approx 0.3010 \approx 0.3010$$

therefore:

$$P(X \leq 0.1) = 1 - 0.3010 P(X \leq 0.1) = 1 - 0.3010 \\ P(X \leq 0.1) \approx 0.6990 \boxed{P(X \leq 0.1) \approx 0.6990}$$

Answer

$$69.90\% \boxed{69.90\%}$$

Interpretation

There is approximately a **69.90%** probability that the next API request arrives within **0.1 seconds**.

Problem 3: Waiting More Than 30 Seconds

A customer-support system receives an average of **4 calls per minute**.

Find the probability that the waiting time for the next call is **more than 30 seconds**.

Step 1: Convert time

$$30 \text{ seconds} = 0.5 \text{ minute}$$

Therefore:

$$x = 0.5$$

and:

$$\lambda = 4$$

Step 2: Use survival probability

$$P(X > x) = e^{-\lambda x}$$

Therefore:

$$P(X > 0.5) = e^{-4(0.5)} = e^{-2} \approx 0.1353$$

Answer

$$13.53\%$$

Interpretation

There is approximately a **13.53%** chance that the support center waits more than 30 seconds for the next call.

Problem 4: Expected Waiting Time for a Customer

A website receives an average of **5 customers per minute**.

Find the expected waiting time until the next customer arrives.

Solution

Given:

$$\lambda = 5 \text{ customers/minute}$$

The expected waiting time is:

$$E(X) = \frac{1}{\lambda}$$

Therefore:

$$E(X) = \frac{1}{5} = 0.2 \text{ minutes}$$

Convert to seconds:

$$0.2 \times 60 = 12$$

Answer

$$12 \text{ seconds}$$

Interpretation

On average, the website waits approximately **12 seconds between customer arrivals**.

Problem 5: Server Failure Analysis

A server experiences an average of **2 failures per month**.

Assume the time until the next failure follows an Exponential distribution.

Find the probability that the server operates for **at least 3 months without failure**.

Given

$$\lambda = 2 \text{ failures/month}$$

We need:

$$P(X > 3)$$

Using:

$$P(X > x) = e^{-\lambda x} \quad P(X > x) = e^{-\lambda x}$$

we get:

$$P(X > 3) = e^{-2(3)} \quad P(X > 3) = e^{-2(3)} = e^{-6} = e^{-6}$$
$$P(X > 3) \approx 0.00248 \quad \boxed{P(X > 3) \approx 0.00248}$$

Answer

$$0.248\% \quad \boxed{0.248\%}$$

Interpretation

There is approximately a **0.248% probability that the server operates for more than 3 months without failure**, under this model.

Problem 6: Machine Breakdown

A manufacturing machine has an average failure rate of:

$$0.05 \text{ failures/hour} \quad 0.05 \text{ failures/hour}$$

Find the probability that the machine does **not fail during the next 10 hours**.

Solution

Given:

$$\lambda = 0.05 \quad \lambda = 0.05 \quad x = 10 \quad x = 10$$

We need:

$$P(X > 10) \quad P(X > 10)$$

Therefore:

$$P(X > 10) = e^{-0.05(10)} \quad P(X > 10) = e^{-0.05(10)} = e^{-0.5} = e^{-0.5}$$
$$P(X > 10) \approx 0.6065 \quad \boxed{P(X > 10) \approx 0.6065}$$

Answer

$$60.65\% \quad \boxed{60.65\%}$$

Interpretation

There is approximately a **60.65% probability** that the machine survives the next 10 hours without failure.

Problem 7: Probability of a Waiting Time Between 2 and 5 Minutes

A call center receives an average of **3 calls per minute**.

Find the probability that the waiting time until the next call is between **2 and 5 minutes**.

Given

$$\lambda=3\backslash\lambda=3$$

We need:

$$P(2<X<5)P(2<X<5)$$

Using:

$$P(a<X<b)=e^{-\lambda a}-e^{-\lambda b}P(a<X<b)=e^{\{-\lambda a\}}-e^{\{-\lambda b\}}$$

Therefore:

$$P(2<X<5)=e^{-3(2)}-e^{-3(5)}P(2<X<5)=e^{\{-3(2)\}}-e^{\{-3(5)\}}=e^{-6}-e^{-15}=e^{\{-6\}}-e^{\{-15\}}$$

Using:

$$e^{-6}\approx0.002479e^{\{-6\}}\backslash\approx0.002479$$

and:

$$e^{-15}\approx0.000000306e^{\{-15\}}\backslash\approx0.000000306$$

Therefore:

$$P(2<X<5)\approx0.0024787P(2<X<5)\backslash\approx0.0024787$$

Answer

$$0.2479\%\backslash\boxed{0.2479\%}$$

Interpretation

There is only about a **0.248% probability that the waiting time is between 2 and 5 minutes.**

This makes sense because the average waiting time is only:

$$13 \text{ minute} = 20 \text{ seconds} \frac{13}{60} \text{ minute} = \frac{13}{60} \times 60 \text{ seconds}$$

Problem 8: Finding the Rate from Mean Waiting Time

A Data Scientist observes that the average waiting time between API requests is **0.25 seconds.**

Find the corresponding Exponential event rate.

Given

$$E(X) = 0.25$$

For an Exponential distribution:

$$E(X) = \frac{1}{\lambda}$$

Therefore:

$$0.25 = \frac{1}{\lambda}$$

Rearranging:

$$\lambda = \frac{1}{0.25} \quad \lambda = 4 \quad \boxed{\lambda = 4}$$

Answer

$$4 \text{ requests per second} \quad \boxed{4 \text{ requests per second}}$$

Data Science Interpretation

If the average inter-arrival time is **0.25 seconds**, the estimated event rate is **4 requests per second.**

Problem 9: Mean and Variance of Waiting Time

An online system receives an average of **10 requests per second**.

Find:

1. Mean waiting time
2. Variance
3. Standard deviation

Given

$$\lambda = 10$$

Mean

$$E(X) = \frac{1}{\lambda} = \frac{1}{10} = 0.1 \text{ seconds}$$

Variance

$$\text{Var}(X) = \frac{1}{\lambda^2} = \frac{1}{10^2} = 0.01 \text{ seconds}^2$$

Standard deviation

$$\sigma = \frac{1}{\lambda} = 0.1 \text{ seconds}$$

Answer

Measure	Value
Mean	0.1 sec
Variance	0.01 sec ²
Standard deviation	0.1 sec

Problem 10: Customer Arrival — Within One Minute

A store receives customers at an average rate of:

$$3 \text{ customers/minute}$$

Find the probability that the next customer arrives within **1 minute**.

Solution

Given:

$$\lambda=3 \text{ } \lambda=3 \text{ } x=1 \text{ } x=1$$

Using:

$$P(X \leq x) = 1 - e^{-\lambda x} \quad P(X \leq 1) = 1 - e^{-3} \quad P(X \leq 1) = 1 - e^{-3}$$

Since:

$$e^{-3} \approx 0.0498 \quad e^{-3} \approx 0.0498$$

we get:

$$P(X \leq 1) = 1 - 0.0498 \quad P(X \leq 1) = 1 - 0.0498 \quad P(X \leq 1) \approx 0.9502 \quad \boxed{P(X \leq 1) \approx 0.9502}$$

Answer

$$95.02\% \quad \boxed{95.02\%}$$

Interpretation

There is approximately a **95.02% probability that the next customer arrives within one minute.**

Problem 11: Finding the Median Waiting Time

An API receives requests at a rate of:

$$\lambda=2 \text{ requests/second} \quad \lambda=2 \text{ requests/second}$$

Find the **median waiting time**.

Solution

The median m satisfies:

$$P(X \leq m) = 0.5 \quad P(X \leq m) = 0.5$$

For an Exponential distribution:

$$1 - e^{-\lambda m} = 0.5 \quad 1 - e^{-\lambda m} = 0.5$$

Therefore:

$$e^{-\lambda m} = 0.5 \quad e^{-\lambda m} = 0.5$$

Taking natural logarithms:

$$-\lambda m = \ln[f_0](0.5) - \lambda m = \ln(0.5)$$

Since:

$$\ln[f_0](0.5) = -\ln[f_0]2 \ln(0.5) = -\ln 2$$

we get:

$$m = \ln[f_0]2 \lambda m = \frac{\ln 2}{\lambda}$$

Therefore:

$$m = 0.69312 m = \frac{0.6931}{2} \quad m \approx 0.3466 \text{ seconds} \quad \boxed{m \approx 0.3466 \text{ seconds}}$$

Answer

$$0.347 \text{ seconds approximately} \quad \boxed{0.347 \text{ seconds approximately}}$$

Important Observation

The mean waiting time is:

$$12 = 0.5 \text{ seconds} \quad \frac{12}{2} = 0.5 \text{ seconds}$$

but the median is:

$$0.347 \text{ seconds} \quad 0.347 \text{ seconds}$$

This happens because the Exponential distribution is **right-skewed**.

Problem 12: 95th Percentile of Waiting Time

An API receives requests at an average rate of:

$$\lambda = 5 \text{ requests/second} \quad \lambda = 5 \text{ requests/second}$$

Find the **95th percentile** of waiting time.

Solution

The 95th percentile x satisfies:

$$P(X \leq x) = 0.95 \quad P(X \leq x) = 0.95$$

Using the Exponential CDF:

$$1 - e^{-5x} = 0.95 \quad 1 - e^{-5x} = 0.95$$

Therefore:

$$e^{-5x} = 0.05 \quad e^{-5x} = 0.05$$

Taking natural logarithms:

$$-5x = \ln(0.05) \quad -5x = \ln(0.05)$$

Thus:

$$x = -\frac{\ln(0.05)}{5} \quad x = -\frac{\ln(0.05)}{5}$$

Since:

$$-\ln(0.05) \approx 2.9957 \quad -\ln(0.05) \approx 2.9957$$

we get:

$$x = \frac{2.9957}{5} \quad x \approx 0.5991 \text{ seconds} \quad \boxed{x \approx 0.5991 \text{ seconds}}$$

Answer

$$95^{\text{th}} \text{ percentile} \approx 0.599 \text{ seconds} \quad \boxed{95^{\text{th}} \text{ percentile} \approx 0.599 \text{ seconds}}$$

Interpretation

Approximately **95% of waiting times are expected to be less than 0.599 seconds** under the model.

Problem 13: Reliability Analysis

A component has an average lifetime of **1000 hours**.

Assume its lifetime follows an Exponential distribution.

Find the probability that the component lasts **more than 1500 hours**.

Step 1: Find λ

The mean is:

$$E(X) = \frac{1}{\lambda} \quad E(X) = \frac{1}{\lambda}$$

Given:

$$E(X) = 1000 \quad E(X) = 1000$$

Therefore:

$$\lambda = \frac{1}{1000} \quad \lambda = \frac{1}{1000}$$

Step 2: Calculate survival probability

We need:

$$P(X > 1500) \quad P(X > 1500)$$

Using:

$$P(X > x) = e^{-\lambda x} \quad P(X > x) = e^{-\lambda x} \quad P(X > 1500) = e^{-\frac{1}{1000}(1500)} = e^{-1.5} = e^{-1.5}$$

$$P(X > 1500) \approx 0.2231 \quad \boxed{P(X > 1500) \approx 0.2231}$$

Answer

$$22.31\% \quad \boxed{22.31\%}$$

Data Science Application

This type of calculation is used in:

- Reliability analytics
- Predictive maintenance
- Equipment lifetime modeling
- Failure-time analysis

Problem 14: Predictive Maintenance

A machine fails on average **once every 20 days**.

Find the probability that it will operate for at least **30 more days** without failure.

Step 1: Determine rate

Mean lifetime:

$$E(X) = 20 \text{ days} \quad E(X) = 20 \text{ days}$$

Therefore:

$$\lambda = 120 = 0.05 \lambda = \frac{1}{20} = 0.05$$

Step 2: Calculate probability

$$P(X > 30) = e^{-0.05(30)} P(X > 30) = e^{-0.05(30)} = e^{-1.5} = e^{-1.5}$$

$$P(X > 30) \approx 0.2231 \boxed{P(X > 30) \approx 0.2231}$$

Answer

$$22.31\% \boxed{22.31\%}$$

Interpretation

There is approximately a **22.31% probability that the machine operates for another 30 days without failure**, assuming the Exponential model.

Problem 15: Real Data Science Problem — Website Traffic

A website receives an average of **120 visitors per hour**.

A Data Scientist wants to analyze the time between successive visitors.

Find:

1. The event rate per minute
2. The average waiting time
3. The probability that the next visitor arrives within 15 seconds
4. The probability that the waiting time exceeds 30 seconds

Solution

Step 1: Convert the rate

Given:

$$120 \text{ visitors/hour} \quad 120 \text{ visitors/hour}$$

There are 60 minutes in an hour.

Therefore:

$$\lambda = \frac{120}{60} = 2 \text{ visitors/minute} \quad \boxed{\lambda = 2 \text{ visitors/minute}}$$

Step 2: Average waiting time

For Exponential distribution:

$$E(X) = \frac{1}{\lambda}$$

Therefore:

$$E(X) = \frac{1}{2} = 0.5 \text{ minute}$$

Convert to seconds:

$$0.5 \times 60 = 30$$

Therefore:

$$E(X) = 30 \text{ seconds} \quad \boxed{E(X) = 30 \text{ seconds}}$$

Step 3: Probability next visitor arrives within 15 seconds

Convert:

$$15 \text{ seconds} = 0.25 \text{ minute}$$

We need:

$$P(X \leq 0.25)$$

Using:

$$P(X \leq x) = 1 - e^{-\lambda x} \quad P(X \leq 0.25) = 1 - e^{-2(0.25)} = 1 - e^{-0.5} = 1 - 0.6065 = 0.3935$$

Therefore:

$$39.35\% \quad \boxed{39.35\%}$$

Step 4: Probability waiting exceeds 30 seconds

Convert:

$$30 \text{ seconds} = 0.5 \text{ minute}$$

We need:

$$P(X > 0.5)$$

Therefore:

$$P(X > 0.5) = e^{-2(0.5)} = e^{-1} = e^{-1}$$

$$P(X > 0.5) \approx 0.3679$$

Therefore:

$$36.79\%$$

Final Answer

Quantity	Result
Arrival rate	2 visitors/minute
Average waiting time	30 seconds
$P(X \leq 15 \text{ sec})$	39.35%
$P(X > 30 \text{ sec})$	36.79%

Data Science Interpretation

A Data Scientist can use this model to understand **inter-arrival times** and support:

- Website capacity planning
 - Traffic modeling
 - Queue analysis
 - Server scaling
 - Customer experience analysis
-

Problem 16: Important Connection — Poisson and Exponential

A server receives an average of **6 requests per minute**.

Question 1

What is the probability of exactly **4 requests in one minute**?

This is a **Poisson** problem.

$X \sim \text{Poisson}(6)$
 $P(X=4) = \frac{e^{-6} 6^4}{4!}$
 $P(X=4) \approx 0.1339$

Therefore:

13.39%

Question 2

What is the probability that the **waiting time for the next request is less than 10 seconds**?

This is an **Exponential** problem.

Convert:

10 sec = $\frac{1}{6}$ minute

Therefore:

$P(X \leq \frac{1}{6}) = 1 - e^{-6(\frac{1}{6})} = 1 - e^{-1} = 1 - e^{-1}$
 $P(X \leq 10 \text{ sec}) \approx 63.21\%$



Poisson vs Exponential

This distinction is **very important in Data Science**.

Distribution	What does it measure?	Example
Poisson	Number of events	Number of website visitors per minute
Exponential	Waiting time between events	Time until next website visitor

Think of it like this:

Suppose customers arrive at a store.

Poisson asks:

"How many customers arrive in the next 10 minutes?"

Exponential asks:

"How long do I have to wait for the next customer?"

Therefore:

Poisson=COUNT\boxed{\text{Poisson}}=\text{COUNT}}

Exponential=WAITING TIME\boxed{\text{Exponential}}=\text{WAITING TIME}}

Important Exponential Distribution Formulas

For:

$X \sim \text{Exponential}(\lambda)$ \boxed{X \sim \text{Exponential}(\lambda)}

PDF

$f(x) = \lambda e^{-\lambda x}, x \geq 0$ \boxed{f(x) = \lambda e^{-\lambda x}, \quad x \geq 0}

CDF

$P(X \leq x) = 1 - e^{-\lambda x}$ \boxed{P(X \leq x) = 1 - e^{-\lambda x}}

Survival probability

$P(X > x) = e^{-\lambda x}$ \boxed{P(X > x) = e^{-\lambda x}}

Interval probability

$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$ \boxed{P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}}

Mean

$E(X) = \frac{1}{\lambda}$ \boxed{E(X) = \frac{1}{\lambda}}

Variance

$\text{Var}(X) = \frac{1}{\lambda^2}$ \boxed{\text{Var}(X) = \frac{1}{\lambda^2}}

Standard deviation

$\sigma = \frac{1}{\lambda}$ \boxed{\sigma = \frac{1}{\lambda}}

Median

$\text{Median} = \frac{\ln 2}{\lambda}$ \boxed{\text{Median} = \frac{\ln 2}{\lambda}}

pp-th percentile

$x_p = \frac{-\ln(1-p)}{\lambda}$ \boxed{x_p = \frac{-\ln(1-p)}{\lambda}}



Memoryless Property

One of the most interesting properties of the Exponential distribution is the **memoryless property**.

It states:

$$P(X > s+t | X > s) = P(X > t) \quad \boxed{P(X > s+t | X > s) = P(X > t)}$$

In simple words:

How long you have already waited does not change the probability distribution of the additional waiting time.

Example

Suppose a server's failure time follows an Exponential distribution.

If it has already operated for 100 hours, the probability of surviving **another 10 hours** is the same as it would have been at the beginning, under the idealized Exponential model.

This property is mathematically useful, although real-world equipment often has aging or wear-out effects, so the Exponential assumption should be validated rather than automatically applied.



Exponential Distribution in Data Science

1. Web Analytics

Time until next visitor \text{Time until next visitor}

2. API Monitoring

Time until next request \text{Time until next request}

3. Queue Analytics

Time until next customer \text{Time until next customer}

4. Predictive Maintenance

Time until machine failure\text{Time until machine failure}

5. Reliability Analysis

Component lifetime\text{Component lifetime}

6. Network Analytics

Time between network events\text{Time between network events}

7. Customer Support

Time until next support request\text{Time until next support request}

8. Event-Driven Systems

Time between incoming events\text{Time between incoming events}



Distribution Series — Quick Summary

You now have:

Distribution	Main Question	Data Science Example
Bernoulli	What happens in one trial?	Customer purchases or not
Binomial	How many successes in n trials?	Purchases among 100 customers
Geometric	How many trials until first success?	Customers until first purchase
Poisson	How many events occur in an interval?	API requests per minute
Uniform	Values equally spread across an interval	Random simulation value
Normal	How are values distributed around a mean?	Measurement/feature values
Exponential	How long until the next event?	Waiting time between API requests

The most important connection to remember:

Poisson $\rightarrow$ How many events?\boxed{ \text{Poisson} $\rightarrow$ How many events?} }

Exponential $\rightarrow$ How much waiting time?\boxed{ \text{Exponential} $\rightarrow$ How much waiting time?} }

That distinction is **extremely useful when selecting probability distributions for real Data Science problems.**